
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become a cornerstone technique for advancing the
2 reasoning capabilities of large language models (LLMs), yet the training dynamics
3 that govern this improvement remain only partially understood. This paper devel-
4 ops a unified account of the RL training process that couples a systematic empirical
5 analysis with a tractable theoretical model of pattern selection. Empirically, a joint
6 reasoning-pattern-level and token-level analysis of the training trajectory surfaces
7 a consistent mechanism: across the tasks and models we examine, the success
8 rate conditioned on an individual reasoning pattern remains approximately stable
9 over training, whereas RL concentrates its updates on a sparse subset of critical
10 tokens and, through them, redistributes probability mass over reasoning patterns
11 in a way that governs overall accuracy. Motivated by these observations, we ab-
12 stract reasoning as a two-stage question-reason-answer process and analyze the
13 training dynamics of the two reward regimes that dominate reasoning-oriented RL:
14 reinforcement learning with verifiable rewards (RLVR) and reinforcement learning
15 from internal feedback (RLIF). Within this abstraction—a tabular pattern-selection
16 policy trained under gradient flow—we characterize, for RLVR, two convergence
17 regimes, one in which the policy converges rapidly to the highest-success-rate
18 reasoning pattern and one in which an entanglement stage sharply delays con-
19 vergence, and we identify the reasoning quality of the base model as the factor
20 that determines which regime obtains. For RLIF, the same analysis explains both
21 why an internal-confidence reward can raise accuracy early in training and why
22 continued optimization may drive the model below its base-model accuracy. Case
23 studies and numerical simulations reproduce each predicted regime, and comple-
24 mentary experiments across additional base models and reasoning benchmarks
25 are consistent with the empirical mechanism. Together, these results sharpen the
26 theoretical understanding of RL in LLM post-training and yield concrete guidance
27 for diagnosing and strengthening reasoning-oriented training.

28 1 Introduction

29 State-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14]
30 now attain remarkable performance across the most demanding classes of logical tasks, spanning
31 mathematics [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) is the technique
32 most directly responsible for this progress, repeatedly demonstrating the capacity to elevate model
33 reasoning to a qualitatively higher level.

34 This success has, in turn, motivated a growing body of work that seeks to explain the mechanisms
35 through which RL improves LLMs, and several complementary accounts have emerged. Comparing
36 pass@ k performance, Yue et al. [42] report that RL-post-trained models rarely surpass their base mod-
37 els, suggesting that RL reorganizes existing reasoning behaviors rather than eliciting fundamentally

new ones. From an entropy perspective, Cui et al. [9], Zhang et al. [45] show that RL-based methods systematically reduce policy entropy, and Cui et al. [9] further establish an empirical link between model performance and policy entropy. Complementing these accounts, Wang et al. [33], Huan et al. [17] find that RL predominantly optimizes a sparse subset of critical tokens. Each of these accounts, however, isolates a single facet of the phenomenon at a fixed snapshot of training; how the token-level, entropy-level, and pattern-level effects arise and interact over the course of optimization—the training *dynamics*—remains comparatively underexplored, both empirically and theoretically. Characterizing these dynamics is the objective of this work.

We approach the RL training process through an analysis that operates jointly at the reasoning-pattern level and the token level across the entire training trajectory. Concretely, we track next-token probability-rank shifts throughout training and, in parallel, apply complementary LLM-based and rule-based extractors to identify and classify the reasoning patterns present in model responses, so that both the distribution over patterns and the pattern-conditioned success rate can be monitored as optimization proceeds. Relative to prior single-snapshot studies [17, 42], this dual-resolution, trajectory-level design yields more direct evidence for a single coherent picture: RL updates concentrate on a sparse subset of critical tokens and, through them, *reshape the distribution over reasoning patterns* that governs model performance, while the success rate conditioned on any individual pattern stays approximately invariant. These regularities recur consistently across the tasks and models we study, and they directly motivate the model we develop next.

Guided by these findings, we introduce a tractable theoretical abstraction that formalizes reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: the model first selects a reasoning pattern conditioned on the question, $\pi(r|q)$, and then produces an answer conditioned on the selected pattern, $\pi(a|r, q)$. This decomposition isolates the stage on which RL primarily acts, and within it we analyze the training dynamics of the two reward types that dominate reasoning-oriented RL: verifiable rewards (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. Under a tabular pattern-selection policy trained by gradient flow, we prove that RLVR optimization drives the policy toward the reasoning pattern r^* with the highest success rate—in agreement with our empirical observations—and we further delineate two distinct convergence regimes: a model with strong initial reasoning quality converges rapidly to the optimal pattern, whereas a weaker one must first traverse an entanglement stage that can drastically slow optimization. For RLIF, the same framework explains both why an internal-confidence reward raises performance early in training and why continued optimization can converge to a state strictly worse than the base model, mirroring the rise-then-collapse behavior we observe empirically. A dedicated suite of case studies and simulations reproduces each of these theoretical predictions.

Our main contributions are as follows.

- **A trajectory-level, dual-resolution characterization of RL training dynamics.** We couple token-level next-token-rank tracking with LLM-based and rule-based reasoning-pattern extraction across the full training trajectory. Compared with prior single-snapshot analyses [17, 42], this design provides more direct trajectory-level evidence that RL updates a sparse subset of critical tokens and thereby *reshapes the distribution over reasoning patterns* that determines performance, while the success rate conditioned on each individual pattern stays approximately invariant throughout training.
- **A two-stage pattern-selection model with convergence analysis for both reward regimes.** We abstract reasoning as a $q \rightarrow r \rightarrow a$ (question-reason-answer) process and, under a tabular pattern-selection policy trained by gradient flow, analyze the induced training dynamics of the two dominant RL paradigms. For RLVR, we prove convergence to the highest-success-rate reasoning pattern and delineate two regimes—efficient convergence for a strong base model versus an entanglement stage that can slow optimization super-exponentially for a weaker one—identifying base-model reasoning quality as the decisive factor. For RLIF, we prove that overall accuracy increases at initialization yet show that continued optimization can converge to a pattern strictly worse than the base model, which accounts for its characteristic rise-then-collapse behavior.
- **Empirical corroboration and practical guidance.** We reproduce every predicted regime through case studies and numerical simulations, and we show that the empirical mechanism transfers across additional base models and reasoning benchmarks. By connecting empirical observation to a tractable theoretical account, our analysis yields concrete, actionable insights for diagnosing and improving RL-based LLM post-training.

94 2 Related Works

95 **Reinforcement Learning for LLMs.** Reinforcement learning has proven remarkably effective for
 96 enhancing large language models (LLMs), both in aligning model outputs with human preferences
 97 [25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. Its central
 98 component is the reward, which conventionally derives from a reward model trained on human-
 99 annotated preference data [26, 25, 2]. More recently, the paradigm has shifted toward rewards that
 100 are cheaper to obtain and directly verifiable, as exemplified by reinforcement learning with verifiable
 101 rewards (RLVR) [30, 40, 14] and reinforcement learning from internal feedback (RLIF) [47, 4]. This
 102 shift has, in turn, catalyzed a family of specialized policy-optimization algorithms, including GRPO
 103 [30], DAPO [41], and VAPO [43].

104 **Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** The empirical success
 105 of RL in LLMs has spurred a substantial literature on its theoretical foundations and mechanistic
 106 interpretation. Early efforts operated largely at the level of algorithm and reward design, addressing
 107 how to construct more informative rewards [28, 18, 38] and how to develop new training algorithms
 108 [37, 11, 21]. As the field has moved toward simpler reward structures such as RLVR and RLIF,
 109 attention has turned to explaining *why* these paradigms are effective, and a range of complementary
 110 accounts has emerged. Shao et al. [29] show that even spurious rewards can enhance reasoning
 111 by unlocking latent skills acquired during pretraining; Agarwal et al. [4] attribute the observed
 112 gains to entropy minimization; and Gandhi et al. [13] isolate specific cognitive behaviors, such
 113 as backtracking, that underpin improved reasoning. Zhao et al. [46] further show that RL fine-
 114 tuning amplifies pre-trained behaviors and drives convergence toward a dominant output format,
 115 a phenomenon that Wang et al. [34], Cui et al. [10] analyze through the lens of entropy collapse.
 116 Each of these accounts examines a single perspective, typically at a fixed point in training. We
 117 complement them by analyzing the RL training *dynamics* through the unifying lens of reasoning-
 118 pattern selection and by grounding that analysis in an explicit pattern-selection model with provable
 119 convergence results—to our knowledge, the first treatment of RL for LLM reasoning from this
 120 dynamical, pattern-selection perspective.

121 3 Preliminaries

122 **Reinforcement Learning for LLMs.** Let π_θ denote a language model with parameters θ , which
 123 serves as the policy subject to optimization. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy
 124 π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$ autoregressively. Post-training casts this generation
 125 problem as a KL-regularized expected-reward maximization:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

126 where π_{ref} denotes the base reference policy and the hyperparameter β weights the KL-divergence
 127 penalty that constrains π_θ from deviating excessively from π_{ref} .

128 The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ admits several concrete instantiations, including a trained reward model
 129 [25] and a rule-based scoring function [30]. We concentrate on the two reward regimes that dominate
 130 contemporary reasoning-oriented RL and that differ precisely in whether supervision originates
 131 outside or inside the model: verifiable rewards (RLVR) [14, 30] and the model’s internal feedback
 132 (RLIF) [47, 4].

133 Under RLVR, the reward $r_\phi(\mathbf{x}, \mathbf{y})$ verifies directly whether the generated answer $\mathbf{y}$ coincides with
 134 the ground-truth answer to question $\mathbf{x}$, giving the canonical binary reward:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

135 In contrast to RLVR, RLIF [47, 4] derives its reward exclusively from intrinsic, model-internal signals
 136 and requires no external verifier. Following the self-certainty formulation of Zhao et al. [47], we
 137 take the RLIF reward to be the average KL divergence between the uniform distribution U over the
 138 vocabulary $\mathcal{V}$ and the model’s next-token distribution, so that a higher reward corresponds to a more
 139 confident (lower-entropy) predictive distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

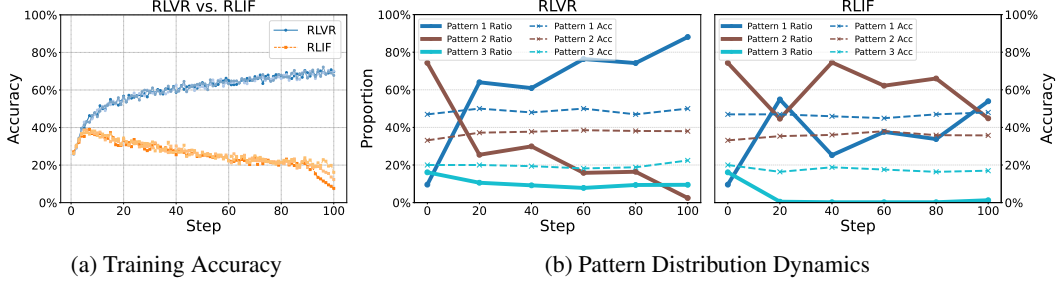


Figure 1: **(a)** Training trajectories of RLVR and RLIF on the MATH dataset for Qwen2.5-3B, with three runs per paradigm obtained under different random seeds. The two methods separate cleanly: RLVR delivers stable, monotone gains, whereas RLIF drives performance up and then back down. **(b)** Reasoning-pattern-level comparison on Number Theory tasks. Solid lines report each pattern’s proportion among all patterns (left axis) and dotted lines its associated success rate (right axis). Under RLVR, probability mass migrates steadily toward the higher-success-rate patterns and away from the lower-success-rate ones, whereas RLIF exhibits unstable pattern dynamics; in both cases the per-pattern success rate stays approximately flat. For readability, patterns are sorted and labeled in descending order of success rate.

140 Because the two regimes share the objective in Eq. 3.1 and differ only in this choice of reward, they
 141 admit a common treatment: throughout the paper we characterize the training dynamics of RLVR
 142 and RLIF within a single empirical and theoretical framework.

143 4 Experimental Exploration for RL Training

144 To characterize the RL training process, we conduct a systematic, multi-resolution study of RL
 145 training under the two reward regimes, structured as a descent through three levels of analysis. We
 146 begin with a trajectory-level overview, which cleanly separates the two paradigms: RLVR yields
 147 steady, monotone improvement, whereas RLIF is unstable and can ultimately degrade performance
 148 below the base model. To explain this divergence, we descend to a reasoning-pattern-level analysis,
 149 which localizes RLVR’s gains in its consistent migration toward patterns with higher success rates
 150 and shows that RLIF fails to specialize in those patterns. Finally, a token-level analysis probes the
 151 substrate of these pattern-level dynamics, showing that they are accompanied by probability-rank
 152 changes in only a small fraction of tokens. Read together—and in contrast to prior single-snapshot
 153 studies [17, 42]—these three levels furnish converging, trajectory-level evidence for a single coherent
 154 account of RL training dynamics: RL concentrates its updates on a sparse subset of critical tokens and,
 155 through them, *reshapes the distribution over reasoning patterns* that determines model performance.

156 4.1 Training Procedure Overview for RLVR and RLIF

157 We first examine the complete training trajectories of RLVR and RLIF under a controlled comparison.
 158 Taking Qwen2.5-3B [39] as the base model and training it on the MATH dataset [16], we hold every
 159 other factor fixed so that the two algorithms differ only in their reward; each paradigm is run under
 160 three random seeds, and we report all resulting trajectories to guard against seed-specific artifacts.

161 As Figure 1a shows, the two paradigms separate sharply and consistently across seeds. RLVR induces
 162 a stable optimization process in which training accuracy improves monotonically and converges
 163 smoothly. RLIF, by contrast, exhibits a characteristic rise-then-collapse profile: accuracy improves
 164 over the early steps but subsequently deteriorates with continued training, in some runs falling
 165 below that of the original base model. This qualitative dichotomy—stable ascent versus transient
 166 gain followed by degradation—is precisely the phenomenon our subsequent fine-grained analysis is
 167 designed to explain.

168 4.2 Reasoning-pattern Level Analysis

169 To explain the pronounced performance disparity between RLVR and RLIF, we conduct a fine-grained
 170 analysis at the reasoning-pattern level. To define and extract patterns, we first collect responses

from the base model and use GPT-4o [19] to cluster them into distinct categories according to their keywords and logical structure, generating a descriptive summary for each pattern, with the extraction and classification prompts run at temperature 0 for reproducibility. Applying these fixed classifications across the training trajectory, we then track, in tandem, the distribution over reasoning patterns and the success rate conditioned on each pattern. The latter is measured by holding the reasoning prefix fixed and completing it with each model, so that any change in measured accuracy is attributable to the pattern distribution rather than to a change in the reasoning process itself.

Concretely, we trace the joint dynamics of pattern proportions and pattern-conditioned success rate under both RLVR and RLIF. Figure 1b presents this analysis on Number Theory tasks drawn from the MATH dataset [16]; consistent results on additional tasks, base models, and scales—including the Geometry dynamics and the endpoint comparisons on Qwen-2.5-7B and QwQ-32B—are deferred to Appendix A. Three findings emerge and recur across these settings:

- Under RLVR, probability mass shifts monotonically toward the reasoning patterns with the highest success rates, which directly accounts for the steady improvement in overall accuracy.
- Under RLIF, the pattern distribution is unstable and does not specialize in the more effective patterns, consistent with the trajectory-level collapse in Figure 1a.
- In both regimes, the success rate conditioned on each individual pattern stays approximately constant throughout training.

Together, these findings localize the performance gap between RLVR and RLIF in their distinct pattern-selection dynamics rather than in any change to the patterns themselves—and the third, in particular, supplies the empirical basis for the fixed-success-rate assumption underlying our theoretical analysis. To identify the substrate that drives this selection, we turn next to a token-level analysis.

4.3 Token-level Analysis

To probe the substrate underlying the reasoning-pattern dynamics observed above, and following the token-level perspective of Huan et al. [17], Yue et al. [42], we conduct a token-level analysis. We sample responses from the base model for questions drawn from the GSM8K [6], MATH [16], and AIME [7] datasets, and, holding these responses fixed, compare the rank of each token—the position of its probability among all vocabulary tokens given the preceding context—between the base and the RL-enhanced models. As summarized in Table 1, individual token ranks are overwhelmingly stable: in most settings fewer than 10% of tokens change rank at all, and the shifting fraction grows only gradually with training step and remains modestly larger for RLIF than for RLVR. The rank metric is deliberately coarse—it registers reordering rather than the magnitude of probability-mass movement—so we read it as indicating that RL does not reorder the response wholesale, rather than as a complete inventory of the underlying probability changes. Even under this conservative reading, the picture is consistent: RL adjusts a sparse set of critical tokens while leaving the ordering of the vast majority of the response intact.

Read together, the three levels of analysis furnish mutually reinforcing, trajectory-level evidence for a single account of RL training dynamics: *RL primarily updates a sparse subset of critical tokens and, through them, reshapes the distribution over reasoning patterns that governs model performance, while the success rate conditioned on each individual pattern remains approximately stable during training.*

Beyond deepening our understanding of RL, these regularities supply the empirical foundation on which we build the pattern-selection model developed in the theoretical analysis that follows.

Table 1: Fraction of tokens whose probability rank changes after RL, reported at successive training steps for fixed base-model responses. Across GSM8K, MATH, and AIME24 and both reward regimes, only a small minority of tokens shift rank, consistent with sparse token-level optimization.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

5 Theoretical Characterization of RL Training Dynamics

We now move from empirical observation to a theoretical account of the RL training process. Guided by the experimental insights of the previous section, we construct a tractable abstraction that decomposes the model’s reasoning procedure into two stages—selecting a reasoning pattern and then deducing an answer from it—and lets RL act on the first, pattern-selection stage. Within this abstraction, and under the simplifying assumptions made explicit below (a tabular policy, gradient-flow updates, and fixed per-pattern success rates), we derive a precise characterization of the training dynamics induced by each reward regime. For RLVR, we isolate two canonical cases with qualitatively distinct optimization behavior; for RLIF, we establish both why internal rewards improve performance early in training and why continued optimization can converge to a model strictly worse than its base. These results are deliberately stylized: they are intended to explain, rather than to reproduce in full detail, the mechanisms observed empirically.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize reasoning as a two-stage generative process: given a question q , the model (1) selects a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that pattern. Pattern r_i is selected with probability $p(r_i|q)$, and each pattern carries its own success rate for producing the correct answer. This decomposition mirrors the empirical picture of the previous section—in which RL redistributes mass over patterns while leaving per-pattern success rates approximately fixed—and it lets us recast the RLIF optimization objective (Eq. 3.3) in pattern-and-answer form¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ the set of output sequences. Given a question q , the language model induces a distribution over output sequences $y \in \mathcal{Y}$ through autoregressive factorization:

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is a function parameterized by θ that predicts the l -th token from the preceding context $y_{<l}$, and the next token is drawn from $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ with logits $z = f_\theta(x, y_{<l})$.

Establishing optimization guarantees for the full nonlinear parameterization is notoriously difficult owing to the complexity of practical models [3, 23]. Following an established line of theoretical work [27, 24, 10, 45], we therefore adopt a *tabular policy* parameterization—a tractable special case of Eq. 5.2 that abstracts away representation learning and non-tabular generalization in exchange for analytical transparency, assigning each output its own trainable logit indexed by the preceding token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to y_{l-1} .

Optimization for RL. To analyze the training dynamics, we pair the tabular policy of Eq. 5.3 with the RL objective $\phi_{\text{RL}}(\theta)$ in Eq. 3.1 and study the policy gradient in the small-learning-rate limit, i.e., the gradient flow, which abstracts the discrete, clipped, finite-sample updates of PPO/GRPO into a continuous idealization:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

Our experiments indicated that the success rate conditioned on each individual reasoning pattern stays approximately stable throughout training. We attribute this stability to a structural asymmetry in the optimization problem: reshaping the mapping from questions to reasoning patterns is considerably easier than reshaping the mapping from a fixed pattern to the final answer. We encode this observation as an idealized assumption:

Assumption 5.1. *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer $\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

Throughout the remainder of this section we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ for the fixed success rate of pattern $\mathbf{r}$. We proceed in two steps: we first derive the optimal policy of the RL objective (Eq.3.1) under the general autoregressive parameterization (Eq.5.2), and then characterize the RLVR and RLIF training dynamics under the tabular policy (Eq. 5.3).

5.2 The Optimal Policy for RL

Our experiments showed that RLVR improves performance steadily by promoting higher-success-rate reasoning patterns, whereas RLIF yields only unstable and ultimately unreliable gains. We trace this contrast to its source by characterizing the optimal policy of the RLVR and RLIF objectives, which already exposes the structural difference between the two rewards before any dynamics are considered:

Proposition 5.2. *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 pins down the optimal solution of each objective and, in doing so, exposes their fundamental divergence. Assuming π_{opt} lies within the general autoregressive policy space—which holds for both LLMs and the tabular policies analyzed below—RLVR raises the selection probability of a reasoning pattern $\mathbf{r}$ precisely when its probability of deducing the correct answer, $p^*(\mathbf{r})$, is large enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$; its reward is thus aligned with accuracy by construction. RLIF, by contrast, rewards only the confidence of the final-answer distribution through $R_{\text{RLIF}}(\mathbf{r})$ and therefore cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Consequently, whereas the RLVR-optimal policy consistently improves performance, the RLIF-optimal policy carries no such guarantee and may settle below the base model—a structural property of the objective, not an artifact of finite training.

Moreover, because β is typically very small in practice (e.g., 0.001) [26, 5, 44], it is instructive—and consistent with common practice—to examine the limit $\beta \rightarrow 0$, in which the optimal policy collapses onto a single pattern:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit the policy becomes deterministic, always selecting the pattern $\mathbf{r}$ with the largest $R(\mathbf{r})$ irrespective of how the reference model initially distributes probability across patterns. The optimal solution alone, however, is silent on the *path* taken to reach it: because the optimization landscape is non-convex, characterizing the endpoint does not reveal how reliably RLVR locates high-reward patterns, nor why RLIF can improve early in training before deteriorating. We address both questions in the following subsections by analyzing the full training dynamics of the two methods under the tabular policy (Eq. 5.3), taking $\beta = 0$ for tractability in line with its small practical value.

5.3 Training Dynamic Analysis for RLVR

Our experiments indicate that the gains from RLVR depend critically on the capacity of the base model. We make this dependence precise by tracking, under gradient flow, the probability assigned to the *optimal reasoning pattern*—the pattern $\mathbf{r}^*$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ —and we show that its dynamics fall into two sharply distinct regimes. In the first, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically and converges to 1. In the second, the trajectory first passes through an *entanglement stage*, during which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ impedes progress toward the optimal one; only after this stage does the trajectory rejoin the first-regime dynamics and converge to $\mathbf{r}^*$. Which regime governs training is determined precisely by the base model’s reasoning quality, as formalized by the two theorems below.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 treats the regime of a sufficiently strong base reference model—one whose overall accuracy already exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under this condition, RLVR drives the model to the optimal reasoning pattern efficiently, at rate $\mathcal{O}(1/\epsilon)$, and attains an overall accuracy approaching the success rate of that optimal pattern.

Practice, however, shows that RLVR optimization can be markedly harder in other settings [44, 36], a difficulty that typically arises when the base reference model is weaker. The second regime captures this case: the model first enters an entanglement stage in which the *suboptimal pattern with the second-highest success rate*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, retards progress toward the optimal pattern.

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(\text{Regime 2}) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 isolates the representative configuration in which only the optimal and suboptimal patterns have success rates exceeding the overall accuracy. We adopt this configuration for analytical tractability; the argument extends, at the cost of a more involved analysis, to the more general setting in which at least two patterns exceed the average success rate.

The central message of Theorem 5.4 is quantitative. RLVR may require T_0 time steps merely to raise the overall success rate above that of the suboptimal pattern—that is, to enter the favorable regime of Theorem 5.3, in which the optimal pattern’s selection probability grows quickly. Whereas the convergence time T_1 of Theorem 5.3 is polynomial in $1/\epsilon$, the transition time T_0 can grow *super-exponentially* in $\gamma_{\pi_{\text{ref}}}$, the ratio between the total selection probability routed through $\mathcal{R}/\{\mathbf{r}'\}$ and that routed through $\mathbf{r}'$. When the base model assigns the optimal pattern only a small selection probability, $\gamma_{\pi_{\text{ref}}}$ becomes large and T_0 prohibitively so, and the model may require an extraordinarily long training horizon before selecting the optimal pattern with appreciable probability. This protracted phase is precisely the entanglement stage, and it explains, within our idealized model, why RLVR from a weak initialization can appear to stall.

Section 6 supplies simulations and case studies that render both regimes explicit.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 already reveals that RLIF, by rewarding only the confidence of the final-answer distribution, cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives and may therefore degrade performance relative to the base model at its fixed point—consistent with our empirical observations. Yet prior studies report that RLIF can improve model performance without any external reward [4, 47]. We reconcile these two facts at the level of dynamics, explaining why RLIF improves a well-trained LLM during the initial stage of training even though its fixed point may be inferior.

We begin from a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Because majority voting [35] reliably improves model performance, a well-trained LLM typically assigns higher probability to the correct answer $\mathbf{a}^*$ than to any incorrect one, so Assumption 5.5 is expected to hold for modern LLMs. Under this assumption we analyze a tractable but representative setting: $|\mathcal{A}| = 2$, success rates drawn uniformly across reasoning paths, and a maximally high-entropy pattern-selection distribution². Analyzing the overall-accuracy dynamics at initialization ($t = 0$) then yields the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. *The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.*
2. *With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.*

The two parts of Theorem 5.6 together account for the full rise-then-collapse profile of RLIF within this idealized setting. The first part establishes that overall accuracy strictly increases at initialization, giving a precise account of RLIF’s early performance gains. As training continues, however, Proposition 5.2 implies that with $\beta = 0$ the policy converges deterministically to the pattern $\mathbf{r}$ maximizing $R_{\text{RLIF}}(\mathbf{r})$, and the second part shows that with probability 0.5 this maximizer coincides with the *least* accurate pattern, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ —so the model converges to the worst reasoning path with probability 50%. The early positive accuracy derivative and the adverse fixed point are thus two facets of a single mechanism: RLIF improves the model at first but, with substantial probability, ultimately drives it toward low-accuracy patterns and below its base-model performance, reproducing the instability observed empirically.

6 Case Studies and Numerical Simulations

To make our theoretical results concrete and to test them under controlled conditions, we complement the analysis with targeted case studies and training simulations within the pattern-selection model. For RLVR, we instantiate the two training regimes of Theorems 5.3 and 5.4, shown in Fig. 2 (upper). For RLIF, we run repeated training simulations in which each reasoning path’s success rate is drawn from a uniform distribution, directly testing the predictions for both the initial training steps and the convergent behavior, shown in Fig. 2 (bottom). In every case the simulations reproduce the predicted behavior.

6.1 Case Studies for RLVR

Section 5.3 identified two distinct regimes in RLVR’s training dynamics: in the first (Eq 5.7), the selection probability of the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ rises monotonically to convergence, whereas in the second (Eq 5.8) the model must traverse an initial entanglement phase and a substantially longer training horizon before rejoining the first-regime dynamics and converging to $\mathbf{r}^*$. Our case studies instantiate both:

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest-entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

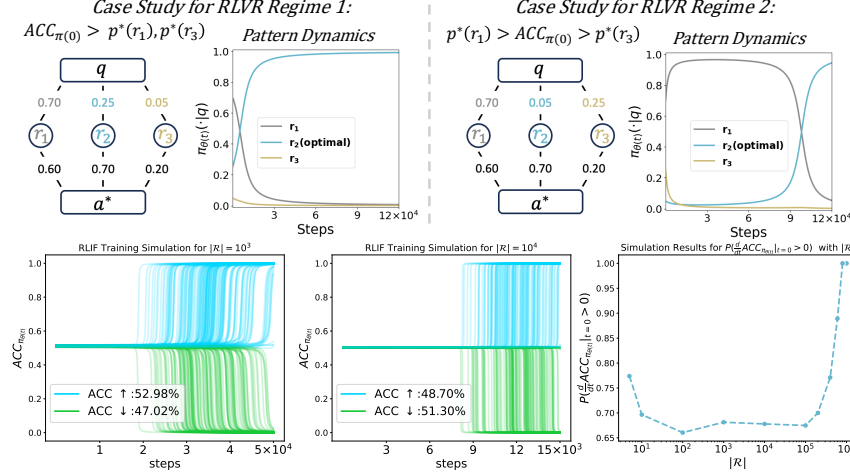


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels instantiate the two predicted RLVR regimes: rapid convergence to the optimal pattern (left) and the entanglement phase that delays it (right). The bottom panels report, for RLIF across varying $|\mathcal{R}|$, the probabilities of converging to an improved versus a degraded state (left) and the probability of an initial performance gain (right). All four panels agree with the predictions of Theorems 5.3, 5.4, and 5.6.

- Fig. 2 (upper left) confirms that when the reference model satisfies Eq 5.7—its overall accuracy exceeding the success rate of every non-optimal pattern—the model converges rapidly to the optimal reasoning pattern, exactly as Theorem 5.3 predicts.
- Fig. 2 (upper right) instantiates the setting of Theorem 5.4: when a suboptimal pattern r' attains a higher success rate than the initial overall accuracy, the model first enters an entanglement phase and must wait out the transition period T_0 (Theorem 5.4) before reaching the rapid-optimization phase of the first regime. Here $\gamma_{\pi_{\text{ref}}}$ (Eq 5.9) is large ($\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$), so the transition time T_0 —which scales as $\gamma_{\pi_{\text{ref}}}^{\pi_{\text{ref}}}$ —becomes prohibitively long and sharply delays convergence. The simulated dynamics exhibit precisely this entanglement stage, in close agreement with the theory.

6.2 Numerical Simulations For RLIF

Section 5.4 predicted that, under Assumption 5.5 and the base-model distribution of Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the model converges to the least accurate reasoning path with probability 50%. Our simulations bear out both predictions. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$ and confirms that the probability of converging to an improved state ($ACC \uparrow$) is approximately equal to that of converging to a degraded state ($ACC \downarrow$), matching the second result of Theorem 5.6. Fig. 2 (bottom right), examining performance after the initial training step, shows that the probability of early improvement approaches 1 as $|\mathcal{R}|$ grows, matching the first result. Jointly, the two panels reproduce the rise-then-collapse dynamics anticipated by the theory.

7 Conclusions and Limitations

This work develops a unified account of reinforcement-learning dynamics in LLMs, pairing a systematic, trajectory-level empirical study with a tractable theoretical model of reasoning-pattern selection. Within a two-stage question-reason-answer abstraction, we analyze the two representative reward mechanisms that dominate reasoning-oriented RL—RLVR and RLIF—and reproduce every prediction through case studies and numerical simulations. Read alongside our empirical findings, the analysis identifies reasoning-pattern selection as a central mechanism through which RL reshapes model performance and singles out base-model reasoning quality as the factor governing convergence.

Several limitations delimit these conclusions and point to concrete next steps. First, the reasoning-pattern categories are obtained through LLM-based extraction and classification; although we fix

the extraction protocol and observe consistent behavior across tasks, models, and scales, the interpretability and reliability of these categories—their stability across prompts, classifiers, and human judgment—warrant more systematic validation, including inter-rater agreement and uncertainty quantification over runs. Second, the theoretical results are established for an idealized model—a tabular pattern-selection policy under gradient flow with $\beta = 0$ and fixed per-pattern success rates—and their extension to realistic settings with KL regularization, clipped and finite-sample PPO/GRPO updates, and non-tabular generalization remains open. Third, the empirical study centers on mathematical reasoning tasks and a specific family of base models, so broadening the analysis across reward designs, hyperparameters, and model families is a natural direction. We view the pattern-selection perspective as a step toward, rather than a complete account of, the mechanisms underlying RL-based reasoning improvement.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

In Section 4.2 we analyzed the reasoning-pattern dynamics on Number Theory tasks sampled from MATH. To verify that this behavior is not task-specific, we repeat the trajectory-level analysis on Geometry tasks (Figure 3). The same picture emerges: under RLVR the model migrates steadily toward the higher-success-rate reasoning patterns, whereas under RLIF pattern selection remains unstable, while the per-pattern success rates stay approximately flat. These results are fully consistent with the empirical and theoretical findings of the main paper.

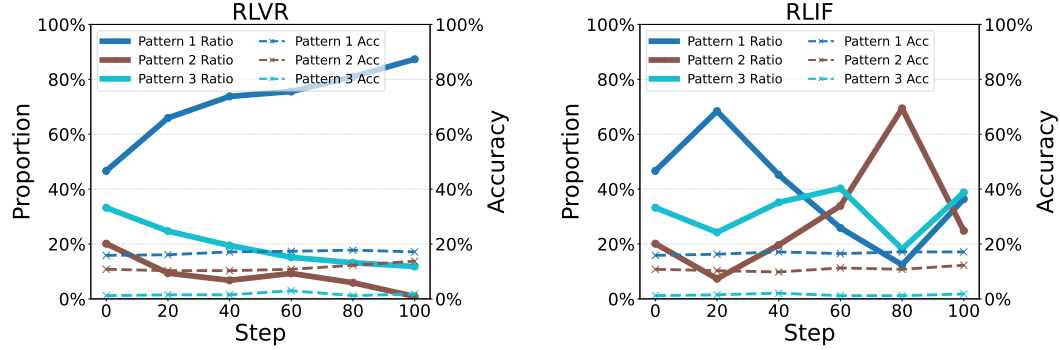


Figure 3: Dynamics of the pattern distribution and corresponding success rate for RL-enhanced models on Geometry tasks. Under RLVR the model converges toward the highest-success-rate pattern (Pattern 1), whereas RLIF remains unstable—mirroring the Number Theory dynamics of the main paper.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To assess whether our findings extend beyond the Qwen2.5-3B setting of the main paper, we conduct endpoint experiments on a second base model (Figure 4). We take Qwen-2.5-7B-Instruct [39] as the base model and compare it against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. The evaluation spans a broad range of mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—using challenging problems sampled from the MATH dataset [16], and extends further to a competition-level problem from AMC23 [1]. Across all of these domains the results echo those of the main paper: RLVR enhancement systematically raises the frequency of the high-success-rate reasoning patterns and suppresses the less effective ones. Detailed task-specific analyses appear in Appendix B.3.

Across this diverse set of tasks, the reasoning-pattern analysis consistently shows that the highest-success-rate patterns become more prevalent after RLVR enhancement, reinforcing the conclusions of Section 4 and Section 5. Notably, in tasks such as Algebra and AMC23-19 (Figure 4) the success rate conditioned on each individual pattern remains stable before and after enhancement; this echoes our earlier observations and provides direct empirical support for Assumption 5.1 of the theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To test whether the mechanism persists at substantially larger scale, we conduct experiments on QwQ-32B [32], which builds on Qwen-2.5-32B-Instruct [39] and strengthens its reasoning ability through RLVR. We compare the reasoning-pattern distributions of the two models on four tasks: the number theory and geometry tasks from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The result is clear: the reasoning patterns of QwQ-32B concentrate on the patterns that are most accurate for Qwen-2.5-32B, consistent with our theoretical predictions. We do not report per-pattern accuracy for QwQ because these models are accessed only through their API, which precludes extracting the chain-of-thought and letting the base model continue generation; consequently the

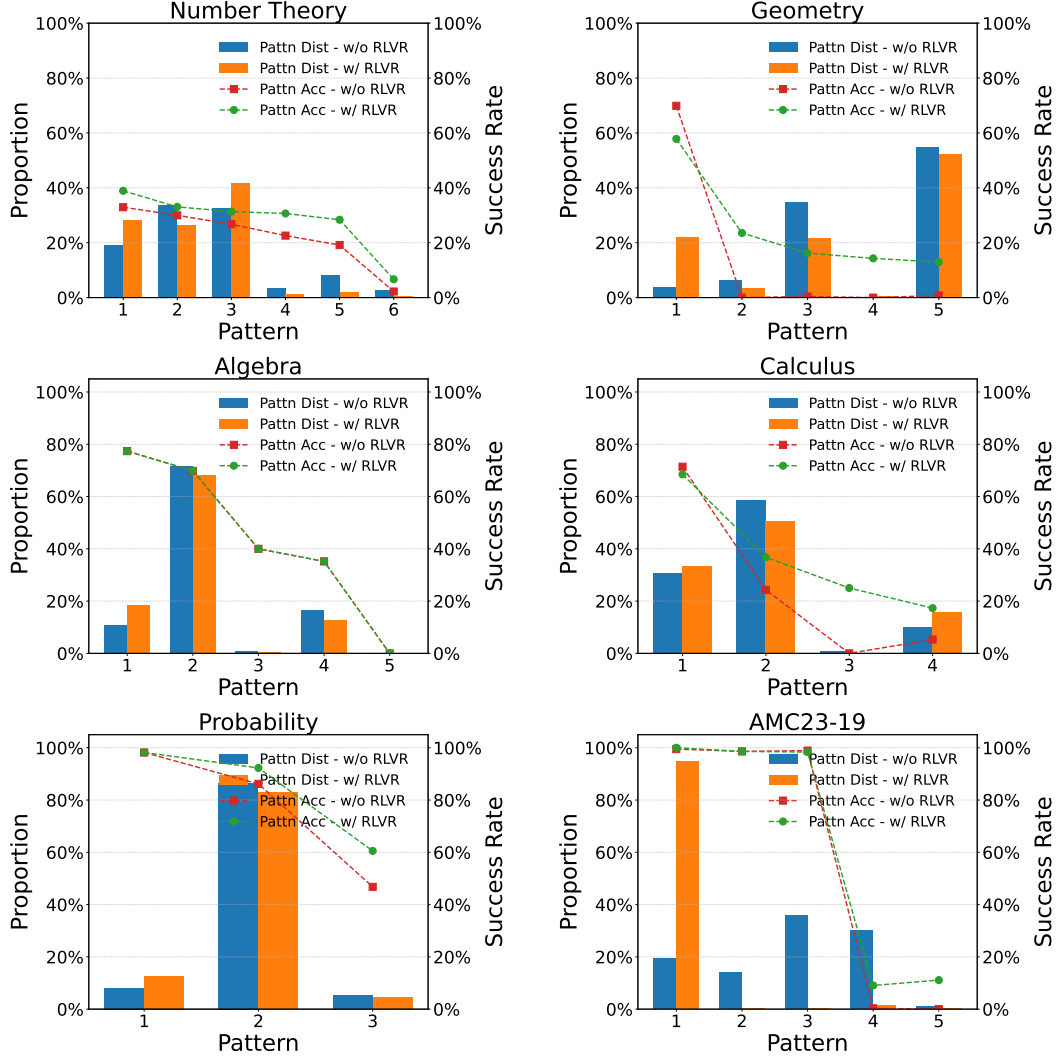


Figure 4: Reasoning-pattern distributions and corresponding success rates for models with and without RLV enhancement across additional tasks, all consistent with our experimental insights and theoretical conclusions. Bars (Pattern Dist) give each pattern’s proportion among all patterns; dots (Pattern Acc) give the associated success rate.

621 pattern-conditioned success rate (as opposed to the model’s overall accuracy) cannot be measured in
622 this setting. See the “Accuracy Analysis” in Appendix B.2 for details.

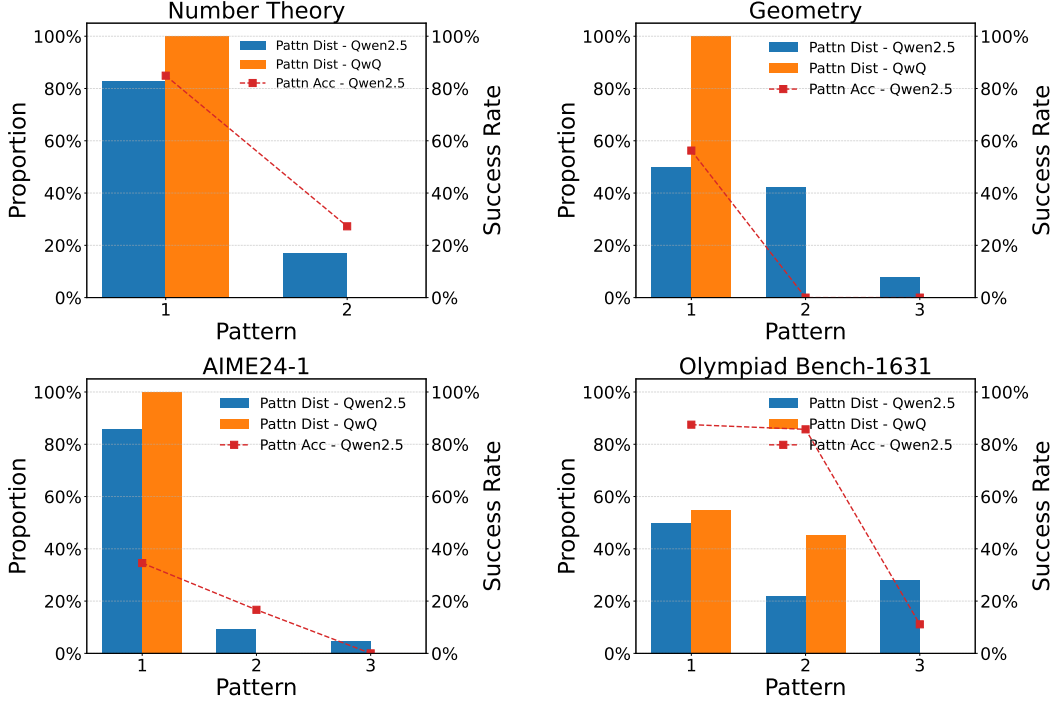


Figure 5: Evaluation on larger models, Qwen-2.5-32B and QwQ-32B. QwQ’s reasoning patterns concentrate on the most accurate pattern of Qwen-2.5, indicating that the pattern-selection mechanism persists at larger scale.

B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We take Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. All experiments use the verl [31] framework with parameter settings identical to the MATH example that verl provides, ensuring a controlled comparison; for RLIF we alter only the reward computation and leave every other setting unchanged. Each configuration is repeated under three distinct random seeds, yielding three independent training curves per method; the qualitative RLVR/RLIF separation reported in the main paper holds across all seeds, indicating that the trajectories are not seed artifacts.

Token-level Analysis: This experiment uses Qwen2.5-3B together with the RL-enhanced models described above. We randomly sample 400 questions from the test sets of GSM8K and MATH respectively, and use the full AIME24 dataset. For each question we first draw an answer from the base model, then concatenate that answer to the question and supply the result as a new prompt to the RL-enhanced model. Using the OpenAI library interface, we compute, for every token in the prompt, its probability and its rank among all candidate tokens at that position. Comparing each token’s rank before and after RL under this teacher-forced protocol yields the proportion of tokens whose rank shifts, which is the quantity we report.

Reasoning-pattern Level Analysis: We conduct two complementary experiments. First, to trace the pattern-distribution dynamics *during* training, we use our own trained models—Qwen2.5-3B and its RL-enhanced counterpart—evaluating every 20 steps. Second, to assess whether the conclusions transfer to other and larger base models, we compare the open-source Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B against their respective base models; for these publicly released checkpoints only the endpoints are available, so we compare the pattern distribution before and after RL rather than tracking its full training-time dynamics.

647 B.2 Reasoning Patterns Analysis Pipeline

648 **Model Inference:** We first randomly sample questions of varying task type (e.g., Geometry,
649 Algebra) and difficulty level (Level 4, Level 5) from datasets including MATH. For each question, the
650 model generates 1024 responses in the pattern-distribution-dynamics experiment, and between 64 and
651 1024 responses—depending on the task category—in the experiment that considers only the before-
652 and-after-RL distribution shift. These responses form the input to the subsequent pattern-extraction
653 and classification stages.

654 **Reasoning Patterns Extraction:** From a sampled subset of base-model responses, we use the
655 GPT-4o [19] API to summarize the categories of reasoning patterns; the full extraction prompt is
656 given in Appendix D.1. To make the extraction deterministic and reproducible, we fix the GPT-4o
657 sampling temperature to 0.

658 **Responses Classification:** We then use GPT-4o, again at temperature 0, to classify every response
659 into one of the identified pattern categories; the full classification prompt is given in Appendix D.2.

660 **Accuracy Analysis:** To separate the effect of the pattern distribution on accuracy from any change
661 in the reasoning process itself, we hold the reasoning process fixed by constructing prompts that
662 concatenate the original question with a partial base-model response, taking that partial response as a
663 faithful realization of a specific reasoning pattern. Concretely, we delete the sentence containing the
664 final answer from the base-model response, append the remaining prefix to the question, and feed the
665 result as a new prompt to the RL-enhanced model, which then completes the answer. The accuracy of
666 this completion defines the success rate of the corresponding reasoning pattern. Because the prefix is
667 held fixed across models, any measured accuracy difference is attributable to the pattern distribution
668 rather than to alterations of the reasoning trajectory.

669 B.3 Details for Each Task

670 Below we detail the experimental procedure for each task, together with its description and represen-
671 tative examples of the extracted reasoning patterns (Table 2). Example prompts for every task are
672 provided in Appendix D.3.

673 **Number Theory:** This task poses coin-distribution problems across multiple bags: the bags
674 initially hold equal numbers of coins, and after additional coins are received and redistributed to
675 preserve equal counts, the total must exceed a specified threshold; the model must determine the
676 minimum initial number of coins per bag. We derive the task from a level-5 MATH problem (id:
677 test/number_theory/1055) [16], generating 32 variants as the evaluation set by adjusting parameters
678 such as the bag count and coin totals. To standardize answer formatting into “boxed{ }” notation
679 without biasing the reasoning approach, we use one-shot prompting with a simple, unrelated example.
680 For evaluation, we draw 20 questions with 4 responses each for reasoning-pattern extraction and
681 sample 64 responses per question; classifying all responses by the extracted patterns then lets us
682 compare the pattern distribution before and after RL.

683 **Geometry:** The geometry task provides the lengths of two sides of a scalene triangle and asks how
684 many distinct integer values (in centimeters) the third side may take. It is derived from a level-4
685 MATH geometry problem (id: test/geometry/1046). Evaluation follows the same pipeline as Number
686 Theory: we construct a synthetic set of 32 questions, draw 20 questions with 4 responses each from
687 the base model (Qwen2.5-7b-Instruct) for reasoning-pattern extraction, and then compare the models
688 with and without RLVR enhancement across all evaluation questions using 64 responses each for
689 pattern analysis.

690 **Algebra:** The algebra task is a can-recycling problem: starting from a given number of old cans,
691 where every n old cans can be recycled into one new can, the model must determine how many cans
692 can ultimately be produced. Its difficulty lies in the leftover cans that cannot be divided evenly at each
693 iterative step and that may later be aggregated for further recycling. The template is derived from a
694 level-4 MATH algebra problem (id: test/algebra/2768). For this and all subsequent supplementary
695 tasks, we adopt exactly the same settings as the preceding tasks, including dataset size and sample
696 counts.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning-pattern examples across the evaluation tasks, together with their characteristic common elements and key words.

Calculus: The calculus task is deliberately concise: given a single-variable trigonometric equation and the domain of its variable, the model must determine the number of solutions. Despite its simplicity, the task probes the model’s basic competence on pre-calculus reasoning. Its prototype is a level-4 MATH pre-calculus problem (id: test/precalculus/1140).

Counting and Probability: We further evaluate the model on a counting-and-probability task. A bin contains white and black balls; given the probability that two randomly drawn balls comprise exactly one white and one black ball, together with the count of balls of one color, the model must compute the minimum number of balls of the other color. This task is derived from a level-4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

AMC23 Question: For this task we use question 19 of AMC23 [1]: we randomly sample 48 responses for reasoning-pattern extraction and a further 1024 responses for the pattern analysis.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We conduct experiments on four tasks. Because these models are accessed only through their API, we randomly select a single question per task, sample 64 responses from each of the two models, and perform reasoning-pattern extraction and classification directly on those responses. Specifically, from the MATH dataset we use question 1055 (number theory) and question 1046 (geometry), and for the remaining two tasks we use problem 1 from AIME24 and question 1631 from OlympiadBench. Representative pattern examples are shown in Table 2.

715 C Deferred Proofs

716 This appendix collects the complete proofs of our main theoretical results—Proposition 5.2 and
717 Theorems 5.3, 5.4, and 5.6—together with the tabular-policy gradient dynamics on which they rely.

718 C.1 Proof of Proposition 5.2

719 *Proof of Proposition 5.2.* In this proof, we utilize the proof techniques in [26], recall that the opti-
720 mization objective of RL is

$$\phi_{\text{RL}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\boldsymbol{\theta}}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

721 Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample
722 $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\boldsymbol{\theta}}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\boldsymbol{\theta}}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\boldsymbol{\theta})$
723 can be written as

$$\phi_{\text{RL}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{y} \sim \pi_{\boldsymbol{\theta}}(\mathbf{y}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\boldsymbol{\theta}}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

724 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\boldsymbol{\theta}) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\boldsymbol{\theta}}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

725 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the
726 partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}}(\mathbf{r}|\mathbf{q}),$$

727 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

728 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

729 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
730 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

731 specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i|\mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

732 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned}\pi_{opt}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})}\end{aligned}$$

733 So we can write $R_{RLIF}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
734 of the theorem. $\square$

735 C.2 Dynamics of Tabular Policy

736 Under the RL objective (Eq 3.1), the tabular policy (Eq 5.3), and Assumption 5.1, the gradient
737 dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt} \theta(t) &= \nabla \phi_{RL}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

738 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
739 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned}\frac{\partial \theta_{r_i,q}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

740 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
741 have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t}$$

742 C.3 Proof of Theorem 5.3

743 The proof of Theorem 5.3 proceeds in three steps: we first show that the overall accuracy is non-
744 decreasing, $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then establish a lower bound on $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$
745 (Eq C.5); and we finally derive the convergence-time bound on t stated in the theorem.

746 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{r_i,q}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

747 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

748 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

749 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
750 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
751 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

752 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
753 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

754 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

755 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

756 This concludes the proof of the theorem. □

757 C.4 Proof of Theorem 5.4

758 The proof of Theorem 5.4 proceeds in three steps: we first establish an upper bound on $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$
759 (Eq C.6); we then analyze the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7); and we conclude by proving
760 that $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$ for all $t > T_0$.

761 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 762 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

763 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

764 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

765 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 766 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

767 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

768 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

769 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

770 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

771 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

772 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

773 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

774 This concludes the proof of the theorem. $\square$

775 C.5 Proof of Theorem 5.6

776 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
777 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

778 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, q))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 779 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, q)) \right). \end{aligned}$$

780 We consider the case $|\mathcal{R}| \rightarrow +\infty$, with the success rates $p^*(\mathbf{r})$ drawn i.i.d. from $U[0, 1]$. By the
 781 central limit theorem, for large $|\mathcal{R}|$ the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ concentrates around its expec-
 782 tation 0.5, with deviation of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We denote this deviation by $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$,
 783 so that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

784 By the Gibbs conditioning principle ([12], Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$ the
 785 empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$ converges
 786 to the density $f(x) = 1 + 12\delta(x - 0.5)$, $x \in [0, 1]$. In this limit we may approximate the summation
 787 over $\mathbf{r}$ by an integral against $f(x)$, and the derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

788 Since $\delta > 0$, the derivative is positive; as $|\mathcal{R}| \rightarrow \infty$ this holds with probability 1, which proves the
 789 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

790 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 791 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

792 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

793 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 794 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 795 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 796 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 797 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 798 $1 - 6\delta$.
- 799 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 800 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

801 This concludes the proof of the second statement.

802

□

803 **D Full Prompts**

804 **D.1 Full Prompt for Reasoning Patterns Extraction**

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge
2. derivation combined with verification
3. independent calculation, suitable for learners starting from the basics,
4. detailed manual derivation, suitable for learners who need to deeply understand each step
5. etc

Or you can classify them into:

1. knowledge call type
2. derivation and verification type
3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

805

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

806

807 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

808

809 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

810

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

811

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

812

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

813

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

814

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

815

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

816

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

817

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

818

Prompt Example for AIME24-Q1

```
<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a  $s$ -kilometer-long walk and stops at a coffee shop afterwards.
When she walks at a constant speed of  $s$  kilometers per hour, the walk takes her 4 hours,
including  $t$  minutes spent in the coffee shop. When she walks  $s+2$  kilometers per hour,
the walk takes her 2 hours and 24 minutes, including  $t$  minutes spent in the coffee shop.
Suppose Aya walks at  $s+\frac{1}{2}$  kilometers per hour. Find the number of minutes the
walk takes her, including the  $t$  minutes spent in the coffee shop.
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim_end>
<lim_start>assistant
```

819

Prompt Example for OlympiadBench-Q1631

```
<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer  $a$ , define a sequence of integers  $x_1, x_2, \dots$  by letting
 $x_1 = a$  and  $x_{n+1} = 2x_n + 1$  for  $n \geq 1$ . Let  $y_n = 2^{x_n} - 1$ . Determine
the largest possible  $k$  such that, for some positive integer  $a$ , the numbers  $y_1, \dots, y_k$ 
are all prime.
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim_end>
<lim_start>assistant
```

820